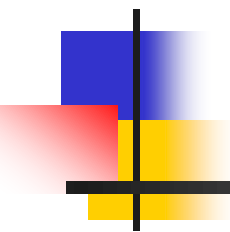
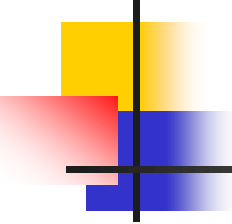


DATA STRUCTURES AND ALGORITHMS



Lecture Notes 2

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ROAD MAP

- **Review Functions**
- Review Classes



Function

- A function is a **group of statements** that has a defined name that is used to refer to that group of statements so that they can be called.
- The groups of statements that make up a function perform a specific task.



Function

- Syntax

```
def functionname(parameter1, parameter2,... ):
    statements
```

- Example

```
def bmiCal(name , weight , height):
    bmi = weight / height ** 2
    print('Hello', name)
    print('Your BMI is' , bmi)
```



Function

```
def Function(n):  
    i = s = 1  
    while s < n:  
        i = i+1  
        s = s+i  
        print("*")
```

```
Function(20)
```

- 1. What's the result of this program?
- 2. How many times the command "print" running?



Classes

- Creating a class is an object-oriented language's ability to manipulate data to solve problems.
- There are two things involved:
 - refer to data or store what data (data members) and
 - what operations can be performed (member functions or methods).



Classes

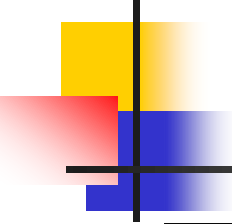
- In python, a new class is created by giving a name and defining a method associated with the definition.

- **Syntax**

class classname:

methods

- After create class, we create instant or object variable to use class.



Classes

```
class Point:
    def __init__(self,x_input ,y_input):
        self.x = x_input
        self.y = y_input

myPoint = Point(3,7)
```

First method is **constructor function**.
In Python, constructor is called with **__init__** (with **two underscores**).



Classes

- The three parameters are (**self**, x_input, y_input).
- **self** is a **special parameter** that will always be used as a reference to the object variable. **It must always be the first parameter.** However, it does not need to be set when called.



Classes

- `self.x` and `self.y` in the constructor are data members of object variable that hold an internal data named `x` and `y`.
- Parameter values are initialized to `x` and `y` when the object variable is created.



Classes

- To create an instance of a Point class (or create an object variable), we do by taking the name of the class and passing it the necessary values (note that **we do not call `__init__` directly**).
- For example:
`myPoint = Point(3,7)`



Classes

- The variable myPoint is an object variable of Class Point (it is a instant of Class Point).

```
class Point:  
    def __init__(self,x_input ,y_input):  
        self.x = x_input  
        self.y = y_input  
    def show(self):  
        print(self.x, ' , ' ,self.y)
```

```
myPoint = Point(3,7)  
myPoint.show()
```